A pre-actuation gate for autonomous agents

NERVA Decision Gating

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$$J(a \mid E, U, C) \geq \theta(a, C)$$

An agent may act only when its evidence clears a bar set by what the action could cost. After a stop, resuming takes more evidence than starting did.

J justification · **a** the proposed action · **E** evidence · **U** uncertainty in the evidence · **C** context: risk, urgency, reversibility · **θ** the bar

Why it exists

Autonomous agents, whether a planner, a learned policy, or a language model, produce confident outputs whether or not the evidence supports them, and act on them at machine speed. Most guardrails inspect the model or audit the outcome. NERVA inspects the entitlement to act: is the evidence sufficient for this action's consequences, and if not, what should happen instead?

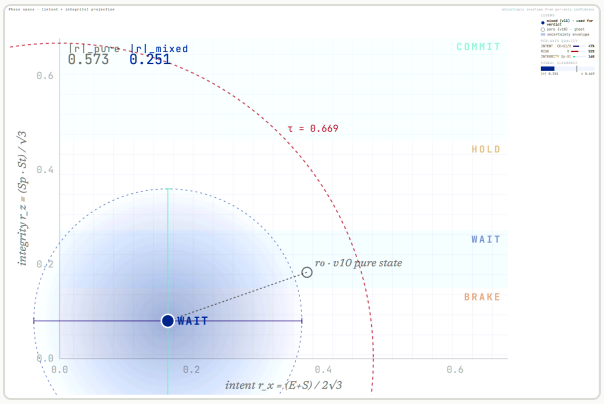
What it does

- Returns one of five verdicts, **COMMIT · HOLD · WAIT · ESCALATE · TOXIC**, each with a receipt listing every input and intermediate value.
- Weights evidence by trust, not agreement. Stale or distrusted sources cannot manufacture confidence, however many of them agree.
- Raises the bar with risk and irreversibility; lowers it when the action can be undone.
- Brakes the instant evidence fails; resumes only when it has clearly recovered. The worst verdict latches, and only an operator releases it.
- Runs locally in tens of microseconds (31 µs median per verdict, pure Python, benchmark in the repo). Spends bandwidth only on the escalate path.

Tested so far

Reference implementation in Python and TypeScript, public on GitHub. Wired into DiMOS, an open robotics stack, so every skill call passes through a verdict before actuation; exercised in replay against recorded sensor traces, not on a physical robot. A four-version falsification series (400 drone-sim trials, predictions documented before testing; version 1 failed on a silent trust-discount flaw and was fixed) and a 20-seed hysteresis evaluation. All simulation. Nothing fielded; nothing independently reproduced yet.

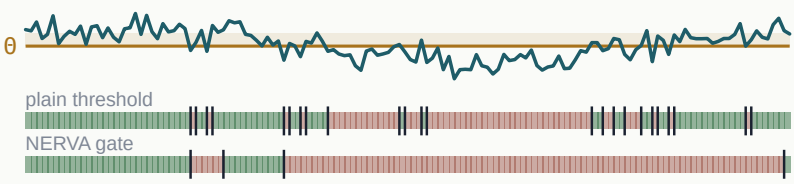
The v11 instrument, as it runs



NERVA is a gate between an agent, whether a language model, a planner, or a robot, and the action it wants to take. The v11 instrument shows one decision. The dashed curve is the bar the action must clear. The ring is where the agent's own numbers put the case for acting; the dot is where NERVA puts it after discounting for how far those numbers can be trusted; the shaded circle is how uncertain that is. The dot falls short of the bar: verdict WAIT, with the reason on the receipt.

How it works

1. **Score.** Five inputs per action: urgency, plan quality, risk, evidence support, environmental stability. Each carries a value and a confidence that reflects how it was obtained.
2. **Discount.** The values combine into a single signal; the confidences shrink it. The implementation keeps this as a density-matrix mixture, used as bookkeeping, not as physics.
3. **Compare and guard.** The signal meets a bar that rises with uncertainty and risk and falls with reversibility. Outside the math, rule guards veto low-integrity high-risk actions, escalate urgent irreversible ones to a human, and brake irreversible actions built on weak inputs.
4. **Verdict, receipt, memory.** The verdict and its receipt go to a ledger. Transitions are asymmetric: an instant brake, a margin and a dwell before resuming.



Same noisy signal, two gates

A plain threshold follows every crossing. NERVA brakes at once and resumes only after the signal clears the bar by a margin and stays there.

24 flips, plain **4** flips, NERVA

Illustrative trace. In a 20-seed replay simulation, hysteresis cut verdict oscillation substantially at a cost of about one second of added recommit delay.

Where it could go

Autonomous systems operating as a team with limited or degraded communications. Each unit needs a local rule for when to act, when to wait, and when to ask. NERVA is that rule. It has not been tested in a multi-agent setting. I am looking for a team building such a system, a testbed to run the gate in, and reviewers who will try to break it.

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